

ADITYA CHUNDURI

MACHINE LEARNING ENGINEER

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EDUCATION

University of Southern California | M.S., Applied Data Science | Dec 2025

SRM Institute of Science and Technology | B.Tech, Computer Science and Engineering (AI/ML) | Jul 2023

SUMMARY

Sole author of a published preprint on LLM agent security evaluation. USC M.S. in Applied Data Science, with computer-vision research, retrieval evaluation, and Python workflows for training-data quality and experiment tracking.

TECHNICAL SKILLS

Programming: Python, SQL, pandas, NumPy.

Vision and evaluation: OpenCV, TensorFlow/Keras, MLflow, Inspect AI, RAG, bootstrap confidence intervals.

Project stack: PyTorch, scikit-learn, Ollama, Gradio, FastAPI, pytest, Docker.

EXPERIENCE

Easley-Dunn Productions, Inc. | Aug 2026 - Present

AI/ML Engineer Intern, Gameplay Analyzer Team | Remote

- Built Python/OpenCV field-registration workflows using yard-line detection and homography estimation.
- Diagnosed spatial-alignment failures across six controlled test categories and documented evidence limits.
- Implemented keypoint projection, per-point residual reporting, spatial checks, and JSONL audit logging.

USC Viterbi School of Engineering | Jun 2024 - Dec 2024

Research Assistant, Computer Vision & Medical Imaging Lab | Los Angeles, CA

- Generated retinal artery/vein masks on RITE and FIVES using RGB-channel differencing and RGB/HSV threshold searches.
- Refined training data through overlap-mask extraction and tracked experiments with MLflow.
- Contributed preprocessing to a team TensorFlow/Keras U-Net pipeline reporting approximately 0.94 AUC and 94% pixel accuracy.

Agent Shield - Independent Research | Oct 2025 - Aug 2026

Independent Researcher, LLM Security and Evaluation

- Developed a six-module Inspect AI framework covering prompt injection, tool poisoning, memory/RAG poisoning, data exfiltration, behavioral drift, and persuasion attacks.
- Benchmarked four LLMs; the published preprint reports a Transparency Rate of 0.000 for three models.

RB Associates | Aug 2021 - Sep 2022

Automation Engineering Intern, Finance and Accounting | Remote

- Automated four internal accounting processes using Python/pandas, schema validation, SQL storage, and rule-based matching.
- Incorporated accountant feedback into human-review workflows; validated reconciliation outputs against manual calculations and documented failure cases.

PROJECTS

Twin - Persona RAG and evaluation | Sep 2026 - Present

github.com/Chunduri-Aditya/twin

- Built a local persona RAG system and frozen 111-item evaluation bank with 95% bootstrap confidence intervals from 1,000 resamples.
- On a synthetic persona, interview-grounded retrieval reduced IPIP-50 response MAE from 0.78 to 0.48 and increased Pearson correlation from 0.52 to 0.78 versus a demographic baseline.
- Compared three embedding indexes on 20 retrieval queries; EmbeddingGemma reached recall@5 of 0.80 versus 0.65 for nomic-embed-text.

MetaLearnML - Model ranking and AutoML tooling

- Meta-feature model ranker with PyTorch/scikit-learn training paths, Neo4j storage, and FastAPI/Streamlit interfaces.
- OpenML benchmark tooling; the September 10, 2026 local audit reported 85 tests passed and 1 skipped.

PUBLICATIONS

- **Beyond Attack Success Rate: Measuring Operator-Facing Transparency in LLM Agent Security.** Sole-authored preprint, Zenodo, 2026. doi.org/10.5281/zenodo.20789431
- **Wind Power Analysis Using Machine Learning in Wind Turbines.** Co-author, IJRASET, 2023. doi.org/10.22214/ijraset.2023.52452